

Predicting MLB draft outcomes for NCAA baseball players: a three-stage XGBoost pipeline with SHAP-based prediction explanations

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Abstract

NCAA players weighing entry to the MLB Draft against returning to school face uncertainty about three quantities: their probability of being drafted, the round in which they will fall, and the signing bonus they can expect. We present a three-stage XGBoost pipeline that jointly models all three using 2021–2025 NCAA data (16,728 player-seasons, 1,438 drafted, 92 features aggregated through our open-source `ncaa_bbStats` package), wrapped in a SHAP-based scouting report layer that converts each prediction into a letter grade, narrative summary, and ranked strengths and concerns. Stage 1 is a balanced binary classifier (89.1% accuracy; PR-AUC 0.687 against a 0.086 baseline). Stage 2 predicts college draft order on the 2025 hold out (Spearman $\rho = 0.710$, MAE 46.5 ranks). Stage 3 predicts signing bonus ($R^2 = 0.454$, $\rho = 0.731$, MAE \$371K). A five-year head-to-head benchmark against MLB Pipeline's top-250 rankings ($n=497$ matched college players, 2021–2025) quantifies the marginal predictive value of in-person scouting beyond publicly available statistics: a mean Spearman gap of 0.224 (model 0.617, Pipeline 0.841). Feature importance converges across all three stages on age, team strength of schedule, offensive production, and strikeout ability.

Keywords

Machine learning, XGBoost, draft prediction, NCAA, sabermetrics

1 Introduction

The MLB Draft, held annually in mid-July, allocates approximately 600 amateur players to professional organizations across 20 rounds (Baseball Almanac 2021–2025). For NCAA players, the decision of whether to enter the draft or return to school carries asymmetric and largely irreversible consequences: a player who enters the draft and signs forfeits remaining college eligibility, while a player who returns risks injury, performance regression, or downward movement in subsequent draft classes. The value of pre-draft information—specifically, calibrated estimates of draft probability, expected draft position, and expected signing bonus—is therefore high, particularly for college juniors who retain leverage by virtue of an additional year of eligibility.

Prior academic work has addressed adjacent questions but, to our knowledge, not this combined question directly. Gerber (2023) models competing risks in the post-draft minor-league pipeline, characterizing the probability of MLB debut and retirement conditional on having been drafted. Penn (2023) quantifies the developmental value of college baseball relative to early professional entry. Seneker (2025) examines signing-bonus negotiation dynamics. Zhao et al. (2025) survey baseball machine learning broadly. Outside the peer-reviewed literature, professional scouting organizations (notably MLB Pipeline) publish pre-draft prospect rankings, and various team-internal and gray-literature models exist; what is absent from the published academic record is a transparent, reproducible pipeline that jointly models draft probability, draft order, and

signing bonus for NCAA players with structured per-player explanations of the predictions.

This paper presents such a pipeline. Our contributions are:

1. A decision-support pipeline that jointly models the three quantities most relevant to an NCAA player's draft-entry decision: a balanced XGBoost binary classifier for draft probability, an XGBoost regressor for college-only draft order, and an XGBoost regressor for signing bonus. We report performance on both balanced and operational (imbalanced) test distributions where applicable.
2. A SHAP-based scouting report layer that converts each player's predictions into a letter grade, one-sentence narrative, and ranked strengths and concerns with feature-level attributions. To our knowledge this is the first published combination of model-based draft prediction with per-player interpretable output suitable for non-technical stakeholders.
3. A five-year head-to-head benchmark against MLB Pipeline's pre-draft top-250 prospect rankings ($n=497$ matched college players, 2021–2025) that quantifies the marginal predictive value of in-person scouting

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beyond publicly available statistics. We measure a mean Spearman gap of 0.224 and discuss the structural component (signability, bonus demands, team strategy) and the substantive component (physical-tools evaluation, pitch quality).

4. An open-source data-collection package, `ncaa_bbStats` (Biggs 2025), that provides reproducible NCAA team statistics (2002–2025), player statistics (2021–2025), and MLB draft results (1965–2025).

Data collection and standardization were performed using the `ncaa_bbStats` package. The first author developed the package as part of this project to address the absence of a unified, reproducible NCAA baseball data interface.

The remainder of the paper is organized as follows. Section 2 situates the work in prior literature. Section 3 describes data sources, scope, features, and missingness. Section 4 details the three predictive stages and the scouting report layer. Section 5 presents results, including the multi-year MLB Pipeline benchmark and case-study scouting reports. Section 6 discusses practical implications, the structural ceiling on draft-order prediction, and limitations. Section 7 outlines future work.

2 Background and related work

2.1 College-to-MLB modeling

The competing-risks analysis of Gerber (2023) provides the closest methodological framework for post-draft career trajectory modeling, treating MLB debut, minor-league retirement, and continued development as competing outcomes. That framework, however, conditions on the player having already been drafted. Penn (2023) characterizes the marginal value of college baseball using a dataset of college-to-pro transitions and finds that younger draftees are systematically undervalued—an empirical regularity that motivates our inclusion of age as a feature and informs our interpretation of feature importance below. Seneker (2025) examines the determinants of signing bonus and team payoff dynamics, providing context for our Stage 3 modeling. None of these works attempts a unified pre-draft prediction pipeline.

2.2 Baseball machine learning

XGBoost (Chen and Guestrin 2016) has been widely adopted in baseball analytics for its ability to handle missing data natively, capture interactions between features without manual specification, and produce interpretable feature importance via gain-based ranking. Huang and Li (2021) apply gradient boosting to MLB win prediction. Zhao et al. (2025) survey machine learning applications in sabermetrics broadly and document the dominance of tree-based ensemble methods for tabular baseball data at the player-season level. Our choice of XGBoost over deep learning reflects three considerations: (i) the sample size (16,728 player-seasons) is small relative to typical deep-learning regimes; (ii) interpretability via feature importance and partial dependence is essential for the decision-support framing; and (iii) the data have no obvious sequential or spatial structure that would favor specialized architectures.

2.3 Rank versus outcome as evaluation targets

Existing baseball draft work typically evaluates against actual draft order or actual draft outcome. We adopt actual draft order as our primary Stage 2 target, both for consistency with prior work and because draft order is the variable players actually observe. We note, however, that actual draft order is bounded above as a predictive target: bonus demands, team strategy, underslot and overslot deals, positional preferences, and medical concerns all create pre-draft signaling that is not statistically predictable from on-field performance. We discuss this structural ceiling in Section 6.

3 Data

3.1 Sources

Data were aggregated through `ncaa_bbStats` (Biggs 2025), an open-source Python package the first author developed for retrieving and standardizing NCAA baseball statistics. The package handles scraping, parsing, fuzzy name normalization, and caching across the upstream sources summarized in Table 1.

Two source details warrant additional comment. First, the Warren Nolan Q1–Q4 splits stratify a team’s record against opponents grouped by RPI quartile, providing a coarse but informative view of how each program performed against varying levels of competition. Second, MLB Stats API fields are populated only for players who appear in MLB’s draft records; they are therefore available exclusively for the drafted subset of our dataset, which is the structural reason physical attributes are used in Stages 2 and 3 but not in Stage 1. Drafted players were matched to their API records via fuzzy name comparison (Section 3) against the draft-class roster returned by the endpoint. MLB Pipeline rankings, which incorporate scouting evaluation, are used only for evaluation and never as training features.

3.2 Scope and rationale

Analysis is restricted to NCAA player-seasons in 2021–2025. This window is motivated by three structural considerations. First, the bat-and-ball equipment regime has been stable since 2015: BBCOR bat standards (introduced 2011) and flat-seam baseballs (introduced 2015) substantially altered offensive baselines in college baseball (Russell 2005), and restricting to the post-2015 era avoids confounding by these changes. Second, the MLB draft itself was reduced from 40 rounds to 20 rounds beginning in 2021 (Gerber 2023); treating pre-2021 and post-2021 draft regimes as comparable would conflate the structural change with player-level effects. Third, FanGraphs player-level coverage at the qualified granularity—meaning a batter needs 3.1 plate appearances per game and a pitcher needs 1 inning pitched per game. Assuming 56 regular-season games, a batter would need 174 plate appearances and a pitcher would need 56 innings pitched—that we require is consistent only from 2021 onward. Together these constraints define the 2021–2025 window as the modern, regime-stable subset of college baseball.

Table 1. Upstream data sources and the fields drawn from each. All sources cover the 2021–2025 seasons.

Source	Fields used
NCAA (2021–2025)	Team-level batting, pitching, and fielding statistics.
FanGraphs (2021–2025)	Player-level traditional and sabermetric statistics (ERA, FIP, K%, wRC+, wOBA, etc.).
Baseball Almanac (2021–2025)	Historical MLB draft results: round and pick number.
Nolan (2021–2025)	Team RPI, strength of schedule, and split-record win-percentages by venue (home, road, neutral), schedule type (conference, non-conference), and opponent quality (Q1–Q4).
Major League Baseball Advanced Media (2021–2025)	Drafted-player attributes: height, weight, primary position, bat side, pitch hand, slot value, and signing bonus.
MLB Pipeline (2021–2025)	Top-250 pre-draft prospect rankings, used for benchmarking only (Section 3.7).

3.3 Dataset construction

The unit of observation is the player-season. After aggregation and deduplication, the dataset comprises 16,728 player-seasons of which 1,438 (8.6%) ended with the player being selected in the MLB draft. Player-to-draft matching was performed via fuzzy name comparison (Python’s `difflib.SequenceMatcher`) against MLB Stats API records, with a 70% similarity threshold. The threshold was chosen by inspecting the empirical similarity distribution, in which confident matches typically score $> 85\%$ and likely mismatches cluster below 60%; we acknowledge that the choice is not optimized against a held-out matched/unmatched ground truth and treat it as a hyperparameter for which sensitivity analysis is future work (Section 7). Five rows (0.3% of the drafted subset) fell below threshold and were dropped, leaving 1,433 drafted player-seasons with confirmed API matches.

We note one additional source of selection bias: drafted players who never appear in our NCAA player-statistics scrape—typically those with fewer than the minimum plate appearances or innings pitched for FanGraphs qualified-noMin coverage—are excluded entirely from the dataset. We do not have a direct count of these players from MLB’s draft records, but their absence biases the drafted subset toward higher-volume contributors. We discuss the implications in Section 6.

3.4 Features

The full feature set comprises 92 variables grouped into three categories (Table 2):

Two variables warrant explicit mention. First, `PE_pct_team` is an engineered feature computed as the ratio of a team’s Pythagorean expectation (Davenport and Woolner 1999) to its actual win percentage, capturing team-level luck or talent-versus-record divergence. Second, `College_Draft_Order` is computed within each year as the rank of a drafted player among college-only draftees (i.e., excluding high-school selections), serving as the Stage 2 target.

The team-context group includes eleven RPI-derived variables sourced from Warren Nolan’s College Baseball RPI pages (Nolan 2021–2025): the team’s own NCAA RPI rating (`rpi_team`), strength of schedule (`SOS_team`), and nine split-record win-percentages (conference, non-conference, home, road, neutral-site, and four quartile-of-opponent buckets Q1–Q4). These features were included because scouts evaluate players against the quality of competition faced; raw conference assignment captures this

only coarsely. As we show in Section 5, the team-level RPI variables emerge as second-tier predictors in Stage 1 and as marginal contributors in Stages 2 and 3.

3.5 Targets

Three targets correspond to the three pipeline stages: a binary `Drafted?` indicator for Stage 1; `College_Draft_Order` (integer rank within year, 1 to ~ 300) for Stage 2; and $\log_{1+x}$ (signing bonus, in millions of dollars) for Stage 3. Signing bonuses in the dataset range from \$1,000 to \$9,250,000, motivating the logarithmic transform to compress the right-skewed tail.

3.6 Missingness

Missingness in the feature matrix is structural rather than random. Pitching features (15 columns) are missing for approximately 67% of player-seasons because those rows correspond to non-pitchers. `TP_team` (team triple plays) is missing for 98% of rows because triple plays are extremely rare at the team-season level. Rather than imputing, we rely on XGBoost’s native missing-value handling, which routes missing values to the optimal child node at each split during training. This is methodologically appropriate here because the missingness itself is informative (a missing ERA tells the model the player is a position player).

3.7 MLB Pipeline benchmark data

For multi-year benchmarking of Stage 2, we scraped MLB Pipeline’s top-250 pre-draft prospect rankings (MLB Pipeline 2021–2025) from mlb.com/mlb/prospects/{year}/draft/ for 2021–2025. The scrape captured 1,250 prospect-rows total (250 per year). Joining against our drafted-player dataset on normalized name yielded 497 matched college players across the five years (99, 97, 108, 96, 97 per year). MLB Pipeline rankings are produced by analysts who combine statistical performance with in-person scouting, physical evaluation, and conversations with team personnel; they thus represent a richer information set than our model uses and serve as a strong baseline.

4 Methods

4.1 Stage 1: Draft probability classifier

With only 8.6% of player-seasons resulting in a draft selection, the classification problem is imbalanced. We address this with random undersampling of the majority class:

Table 2. Feature groups used across the three pipeline stages. Physical attributes are available only for drafted players (MLB Stats API), so they are used in Stages 2 and 3 only.

Group	Count	Examples
Player batting	17	AVG, OPS, wRC+, wOBA, K%, BB%, ISO
Player pitching	16	ERA, WHIP, FIP, K/9, BB/9, K-BB%, LOB%
Team batting	16	BA, SLG, OBP, R/G, SB/G
Team pitching/fielding	18	ERA, WHIP, FPCT, K/9, SHO
Team context	25	W/L, WPCT, conf. RPI, PE, PE_pct, RPI, SOS, splits WP
Physical (drafted only)	5	height, weight, bat side, throw side, pos
Total	92	

we sample 1,438 non-drafted player-seasons (matching the drafted count) and train on the resulting balanced set of 2,876 observations. An 80/20 stratified train/test split preserves the 50/50 balance in both partitions.

The XGBoost classifier uses 500 trees with learning rate 0.05, maximum depth 6, subsample 0.8, and column subsample 0.8. Regularization is provided implicitly by the subsampling and by limiting tree depth.

We considered two alternative strategies for handling imbalance: (i) training on the full imbalanced data with `scale_pos_weight` set to the negative-to-positive class ratio (~ 10.6), and (ii) SMOTE oversampling of the minority class (Chawla et al. 2002). We chose undersampling for two reasons. First, evaluating a model on imbalanced test data without rebalancing inflates accuracy by allowing the model to exploit the majority-class base rate; balanced evaluation forces the model to demonstrate genuine signal. Second, the drafted subset (1,438 players) is large enough that information loss from undersampling is modest, while undersampling avoids the synthetic-feature artifacts SMOTE can introduce.

Evaluation on operational distribution. The balanced test set is appropriate for measuring whether the model has learned a discriminative function, but it does not reflect deployment, where the model would see player-seasons drawn from the natural 8.6% positive rate. We therefore report two complementary evaluations: per-class precision, recall, and F1 on the balanced held-out set (Table 3), and threshold-independent area under the precision-recall curve (PR-AUC) on a held-out subset of the full imbalanced player-season pool (Section 5). PR-AUC is appropriate for skewed-class problems because, unlike ROC-AUC, it does not give credit for correctly classifying the abundant negative class (Saito and Rehmsmeier 2015).

4.2 Stage 2: College draft order regressor

Stage 2 operates on drafted players only ($n=1,433$ after the 70% name-match filter). For each year, we compute `CollegeDraftOrder` by sorting drafted players by overall pick number and assigning a within-year rank to college-only draftees (high-school selections are excluded so that the rank is over a homogeneous population).

Single-year hold-out. The primary evaluation trains on 2021–2024 ($n \approx 1,159$) and tests on 2025 ($n=274$). The XGBoost regressor uses 1,000 trees with early stopping (patience 50 rounds, monitoring validation MAE), learning rate 0.03, maximum depth 4, minimum child weight 5, L_1 regularization 1.0, L_2 regularization 5.0, subsample 0.8, and

column subsample 0.7. Hyperparameters were chosen by manual grid search on 2021–2023 training with 2024 as validation; we report the 2025 hold-out as the unbiased test set.

Multi-year head-to-head. For benchmarking against MLB Pipeline across 2021–2025, we use leave-one-year-out cross-validation: for each year $y \in \{2021, 2022, 2023, 2024, 2025\}$, we train on the other four years and predict on y , yielding 1,433 out-of-fold predictions in total. We use a single hyperparameter configuration (identical to the single-year hold-out, tuned on the 2021–2023/2024 split) across all five LOYO folds. This is a deliberate choice: re-tuning hyperparameters per fold would introduce additional researcher degrees of freedom and is the kind of nested-CV procedure that risks overfitting to the LOYO evaluation itself. The cost of this choice is that the 2025 hyperparameters may be slightly suboptimal for the 2021 and 2022 folds; we view this as a conservative bias against the model in the head-to-head.

We also evaluated a percentage-target variant predicting `CollegeDraftPct` (normalized within-year rank in $[0, 1]$) to enable comparison across years with different draftee counts; the raw-order model achieved slightly stronger Spearman (0.710 vs. 0.698) and we report it as primary.

4.3 Stage 3: Signing bonus regressor

Stage 3 also operates on the drafted subset, restricted to players with MLB Stats API signing-bonus data ($n=1,400$; 97.7% of drafted-with-API-match). Training is on 2021–2024 with the 2025 hold-out ($n=273$). Hyperparameters match Stage 2.

A critical methodological consideration is leakage prevention. The following features are explicitly excluded from the Stage 3 feature set:

- `api_pickValue` (the league-assigned slot value, mechanically determined by pick number);
- `Pick` and `Round` (the actual pick number and round, which are deterministic proxies for the signing-bonus distribution);
- `CollegeDraftOrder` (the Stage 2 target, which would create cascading leakage).

The remaining feature set is identical to Stage 2.

4.4 Evaluation protocol

For Stage 1, we report accuracy, precision, recall, and F1 per class on the balanced test set, plus the gain-based top-20 feature importance ranking.

For Stage 2 single-year, we report Spearman ρ with a 95% confidence interval computed via the Fisher z -transform, MAE on rank, and head-to-head performance against MLB Pipeline on the 2025 matched subset. For Stage 2 multi-year, we report per-year Spearman and MAE for both the model and MLB Pipeline across all five years ($n=497$ total matched).

For Stage 3, we report R^2 , MAE in dollars (back-transformed from the log target), and Spearman ρ .

For all three stages, we report feature importance and 1D partial dependence plots for the top features; for the classifier we also report 2D interaction plots for the strongest pairs.

4.5 Player-facing scouting report layer

The three predictive models produce three numerical outputs for each player (draft probability, predicted college draft order, predicted signing bonus). For these numbers to support individual player decision-making, they must be paired with an interpretable account of *why* the model produced them. We wrap the trained models in a scouting report layer that emits, for any player or custom statistical profile:

- A **letter grade** for the draft probability and predicted draft rank, calibrated against the training distribution. Draft grades use absolute thresholds ($P[\text{draft}] \geq 90\% = A+$, $\geq 80\% = A$, descending through D–/F at $< 1\%$); rank grades use percentile bins of the training set’s predicted-rank distribution.
- A **one-sentence narrative** (“*the model sees a top profile (~pick 17). Driven by SO (180 vs. 55 median). Main red flag: Age (20 vs. 21 median)*”), generated from a fixed template populated with the grade band, top strength, and top concern.
- A ranked list of the player’s **top five strengths** and **top five concerns**, with each entry showing the raw feature value, the training-median value, the feature’s contribution to the draft-probability prediction in percentage points, and a cross-check importance $\times z$ -score.
- A **rank-model cross-check** naming the three features that most improve the predicted draft position and the three that most hurt it.

Feature contributions use two parallel methods. The primary attribution is the SHAP TreeExplainer (Lundberg and Lee 2017), which produces exact per-feature contributions to each tree-ensemble prediction. For the draft-probability classifier, the contribution is expressed in percentage points on the predicted probability (computed by inverse-logit transforming the log-odds contribution); for the rank model, in expected pick-number displacement. A secondary diagnostic computes importance $\times z$ -score, defined for each feature f and player as

$$\text{imp}_f \times \frac{x_f - \tilde{x}_f}{\sigma_f},$$

where imp_f is the model’s normalized gain importance, $\tilde{x}_f$ is the training-set median, and σ_f is the training-set

Table 3. Stage 1 classifier performance on the balanced held-out test set ($n=576$). The model captures genuine signal in both directions; the modest precision-recall asymmetry favors recall of the drafted class, which is appropriate for decision support.

Class	Precision	Recall	F1	n
Not drafted	0.934	0.840	0.885	288
Drafted	0.855	0.941	0.896	288
Accuracy: 89.1%				

standard deviation. This serves as a model-free sanity check that does not require SHAP and is used when the SHAP and importance $\times z$ -score signs disagree to flag potentially unstable attributions.

We impose a suppression rule for downstream predictions: when the classifier outputs $P[\text{draft}] < 25\%$, the predicted rank and signing bonus are hidden in the report. The rank and bonus models are trained only on drafted players, so applying them to a clearly-undrafted profile constitutes extrapolation outside the training distribution. Suppression is preferable to displaying unreliable numbers.

The scouting report serves two operational goals. First, for an individual NCAA player, it converts an opaque model prediction into a structured account of which on-field statistics are pulling their draft stock up and which are pulling it down. Second, for downstream use cases (e.g., agent advisory or coach feedback), it provides a paragraph-level summary suitable for sharing with non-technical stakeholders without exposing the underlying feature matrix.

5 Results

5.1 Stage 1: Draft probability classifier

The balanced classifier achieves 89.1% accuracy on the held-out test set, with strong per-class performance (Table 3). The drafted class achieves recall of 0.941 with precision of 0.855 (F1 0.896); the not-drafted class achieves precision of 0.934 with recall of 0.840 (F1 0.885). The model is therefore slightly biased toward classifying borderline players as drafted, which is appropriate for a decision-support system whose downstream rank and bonus predictions are themselves suppressed when draft probability is low.

Performance on the operational imbalanced distribution. The 89.1% balanced accuracy reported above is the appropriate metric for evaluating whether the model has learned a discriminative function, but it is not the metric that matters for deployment. On the operational distribution (8.6% drafted), the same per-class recall values would yield a substantially lower precision on the positive class because false positives accumulate from the much larger negative pool. To characterize operational performance, we constructed a held-out evaluation set at the natural 8.6% positive rate ($n=3,348$) by combining the 288 drafted players already in the balanced held-out test set with 3,060 not-drafted players randomly sampled from the population the model never saw during training (i.e., the not-drafted players excluded from the balanced training+test pool). The model achieves PR-AUC of 0.687 on this imbalanced test set

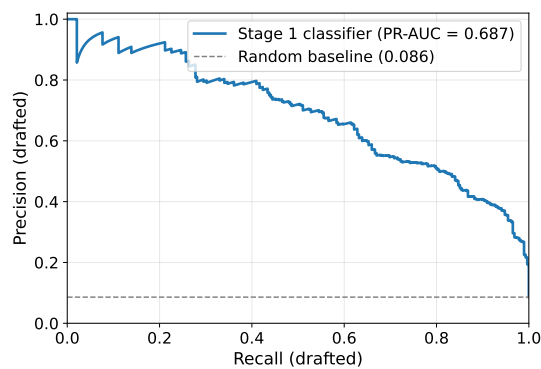


Figure 1. Stage 1 precision-recall curve on the operational imbalanced test set ($n=3,348$, 8.6% positive rate). The classifier (PR-AUC = 0.687) substantially outperforms the random baseline (0.086) across the full recall range. The curve makes the precision-recall tradeoff at different operating points explicit: deployments requiring higher precision can raise the probability threshold to operate in the high-precision/low-recall region (e.g., precision ≈ 0.80 at recall ≈ 0.40), while the default 0.5 threshold corresponds to the high-recall regime (precision 0.374, recall 0.941).

against a random baseline of 0.086—an approximately eight-fold lift over base rate—and ROC-AUC of 0.956. Figure 1 shows the full precision-recall curve, which sits comfortably above the random baseline across the entire recall range and makes the precision-recall tradeoff at different operating points visually explicit. The substantial gap between the two AUC measures is diagnostic: the model ranks drafted players very near the top of its probability ordering (hence the high ROC-AUC) but absolute precision is constrained by the size of the negative pool. At the default 0.5 probability threshold, operational precision on the drafted class is 0.374 with recall 0.941: when the model flags a player-season as draftable, it is correct 37% of the time and captures 94% of actual draftees. Lowering the threshold to 0.3 produces only a marginal recall gain (to 0.965) but degrades precision substantially (to 0.328), indicating that 0.5 is already near the high-recall regime of the model’s predicted-probability distribution; deployments requiring higher precision should raise the threshold rather than lower it. Threshold selection in practice should be governed by the relative cost of false positives (recommending draft entry to a player who will not be drafted) versus false negatives (telling a draft-quality player they are unlikely to be selected), which is application-dependent.

Feature importance (Figure 2) is led by age (gain 18.5) and the team RPI rating (`rpi_team`, gain 13.5), with offensive production (`wRC`, `SLG`) and strikeouts (`SO`, `K%`) filling the rest of the top tier. The prominence of `rpi_team` and `SOS_team` confirms that the model interprets a player’s individual statistics in relation to the quality of competition faced—a player putting up strong numbers in a high-RPI conference is more strongly signaled as drafted than the same statistics in a weaker conference. The team’s own RPI is a substantially richer signal than the conference-level alternative: `conf_rank_team` (the team’s standing within its conference) does not appear in the top 10 once `rpi_team` is available.

Partial dependence plots (Figure 3, top row) show the expected monotonic shape for offensive features: more production, higher draft probability. The age PDP is non-monotonic: draft probability rises through age 20 and peaks at 21 before declining, consistent with the empirical pattern that teams systematically prefer college juniors over fifth-year seniors due to leverage considerations (Penn 2023). The two-dimensional age $\times$ `wRC` PDP (bottom-left of Figure 3) shows a strong interaction: high offensive production is necessary but not sufficient for old players, while young players with even moderate production retain high draft probability.

Qualitative inspection of the classifier. Figures 4 and 5 visualize the relationship between top pitching features and draft status using parallel coordinates. The actual-status plot shows that drafted pitchers (blue) concentrate in the high strikeout, high-`k/9`, high-innings region of feature space, while undrafted pitchers (red) span a much wider range. The predicted-status plot shows the same structure recovered by the classifier on held-out test data, confirming that the model has learned the qualitative separation reflected in the feature importance ranking.

5.2 Stage 2: Draft order, 2025 hold-out

The Stage 2 model trained on 2021–2024 achieves Spearman $\rho = 0.710$ (95% CI [0.645, 0.764] via Fisher z) on the 2025 hold-out ($n=274$), with MAE of 46.5 ranks. Figure 6 shows predicted versus actual draft order; the model captures the broad ordering well, with the strongest accuracy among the top-20 actual picks.

Inspection of the top-ranked 2025 college players (Table 4) shows tight agreement near the top of the board: Anderson (predicted 2, actual 1), Doyle (predicted 4, actual 2), Arnold (predicted 3, actual 4). The largest mid-first-round misses—Bodine (predicted 84, actual 12), Summerhill (predicted 81, actual 20), and Quick (predicted 63, actual 16)—share a profile of moderate statistical production paired with strong physical tools and scouting attention, exactly the features our stats-only model cannot observe.

5.3 Stage 2: Multi-year head-to-head vs. MLB Pipeline

Using leave-one-year-out predictions, we compare the stats-only XGBoost model against MLB Pipeline’s pre-draft top-250 prospect rankings (MLB Pipeline 2021–2025) across all five years (Table 5). Across 497 matched college players, the model achieves a mean per-year Spearman of 0.617 with mean MAE of 19.3 ranks, while MLB Pipeline achieves 0.841 and 12.0 respectively.

Three observations bear discussion. First, the model exhibits stable per-year performance (Spearman 0.605–0.688) across 2022–2025, suggesting the underlying signal is consistent across recent draft regimes. Second, 2021 is an outlier: the model’s Spearman drops to 0.468, while Pipeline’s holds at 0.822. We hypothesize that 2021 statistics, drawn from the first full post-pandemic NCAA season, contained reduced signal due to truncated 2020 schedules, roster-eligibility changes, and incomplete RPI accounting; scouts (whose Pipeline rankings incorporate longer histories and in-person evaluation) were less affected. We do not

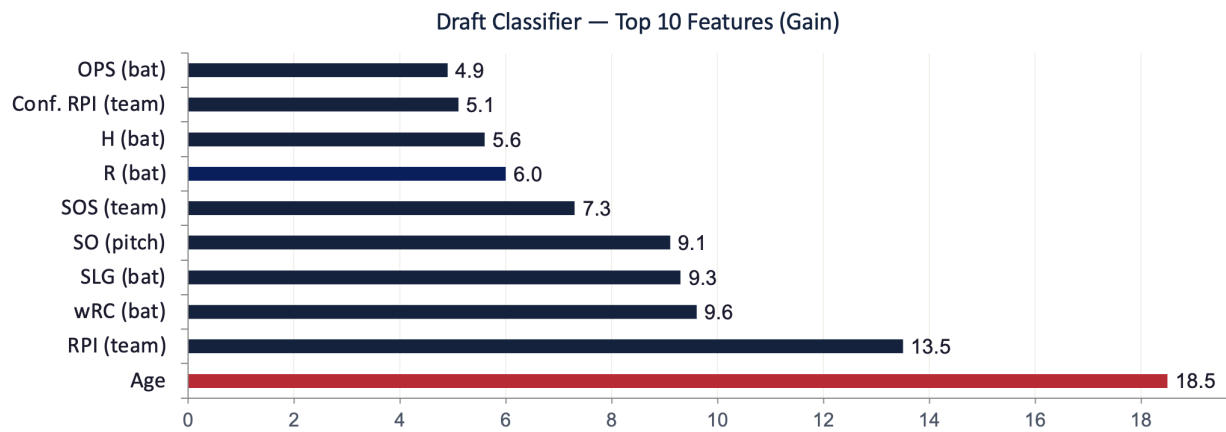


Figure 2. Top-10 features by gain for the Stage 1 classifier. Age leads, followed by team RPI; offensive production (wRC, SLG) and strikeout metrics (SO) form the second tier; team strength-of-schedule (SOS) and conference RPI provide the remaining signal.

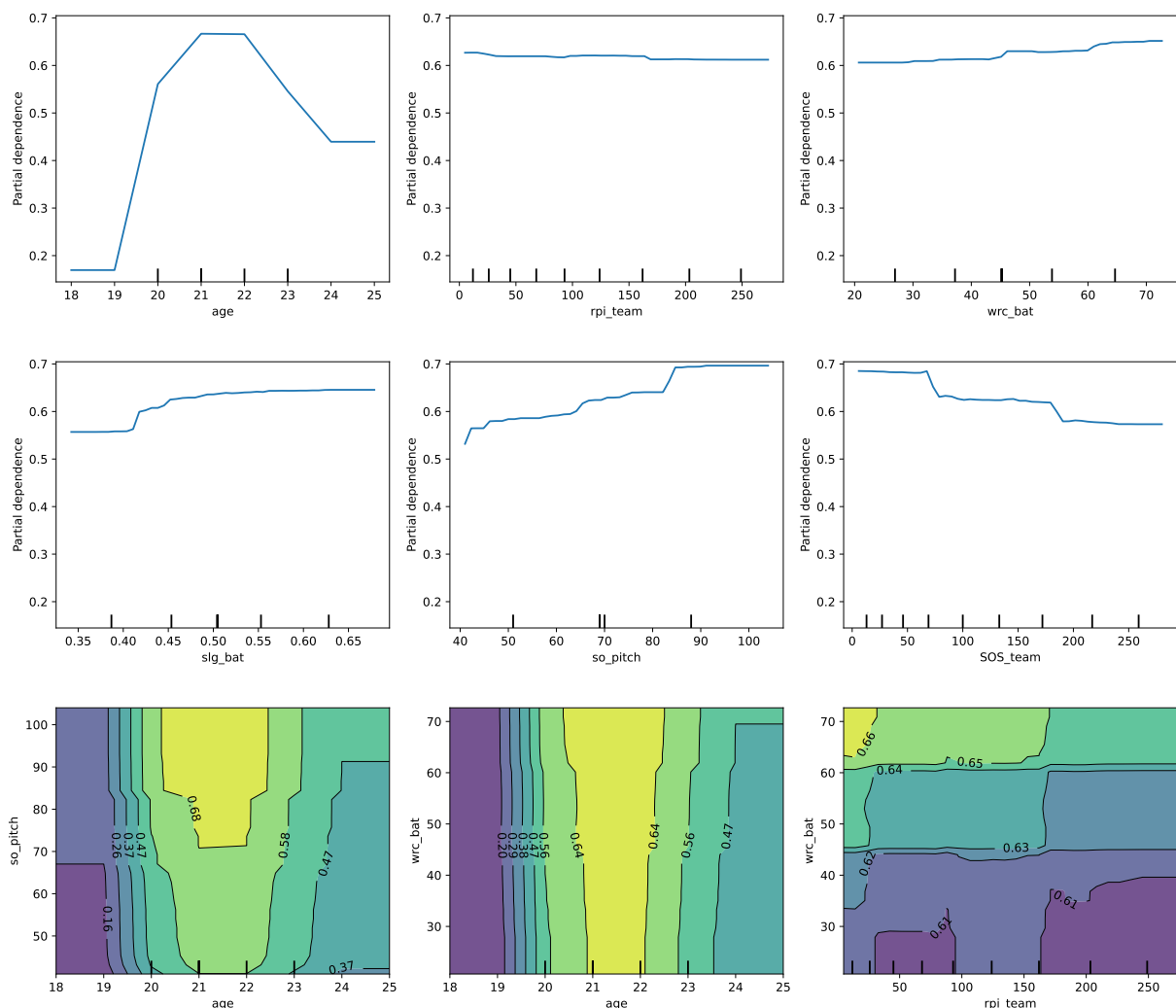


Figure 3. Partial dependence plots for the Stage 1 classifier. Top two rows: 1D PDPs for the six highest-gain features (age, team RPI, wRC, SLG, strikeouts, and SOS). Bottom row: 2D interaction PDPs for the strongest feature pairs (age $\times$ strikeouts, age $\times$ wRC, and team RPI $\times$ wRC). Offensive features and strikeouts show the expected monotonic relationship; age peaks at 21 and declines; the strongest joint effects appear where above-average individual production combines with favorable age or team context.

have the data to confirm this hypothesis definitively. Third, the persistent ~ 0.22 gap between model and Pipeline mean Spearman is consistent with prior expectations: Pipeline rankings incorporate physical-tools evaluation, scouting

reports, and team-internal signaling that our model does not access.

We emphasize that the gap to actual draft order is bounded above by structural factors (Section 6): even a perfect-information ranker would fall short of $\rho = 1.0$ against actual

figures/pitching_pcp_drafted.pdf

Figure 4. Parallel coordinates plot of top pitching features by actual draft status. Drafted pitchers (blue) cluster at higher strikeout totals, higher k/9, and higher innings pitched than undrafted pitchers (red).

pick number, because actual pick number is influenced by bonus demands, signability, team strategy, positional preferences, and medical concerns that are not predictable from statistical performance alone.

5.4 Case study: what the model cannot see

To make the information gap concrete, we examine a single 2025 player in detail: Kade Anderson, the eventual No. 1 overall pick in the 2025 MLB Draft and a left-handed pitcher at Louisiana State University (LSU). Our Stage 2 model ranked Anderson 2nd among 2025 college draftees; MLB Pipeline ranked him 1st. The model's ranking is competitive with the scouting consensus at the very top of the board.

Anderson's 2025 statistical line, the basis for our model's prediction, was excellent: he led NCAA Division I with 180 strikeouts in 119 innings, finished third nationally with 12 victories, and recorded a sub-3.50 ERA. His K% and K/9 metrics, the strikeout-related features that anchor the

Stage 1 importance ranking (Figure 2), placed him among the top handful of NCAA pitchers. The model captured this performance and ranked him accordingly at the top of the 2025 college class.

What the model did not capture, and what MLB Pipeline did, is the qualitative scouting profile that consolidated Anderson's position as the presumptive No. 1 overall pick. Pipeline's pre-draft writeup ([MLB Pipeline 2021-2025](#)) documents several facts not encoded in any of our 92 features:

- **Injury and recovery context.** Anderson underwent Tommy John surgery in April 2022 and missed his Louisiana high school senior season the following spring. LSU eased him into 2024 as a midweek starter and weekend reliever before he became a full-time starter in 2025. Our model observes only the 2025 season's statistics; it cannot see that this represented a single dominant year after a major elbow

figures/pitching_pcp_predicted.pdf

Figure 5. Parallel coordinates plot of the same features colored by the classifier's predicted draft probability on the held-out test set. The model recovers the same qualitative separation between drafted and undrafted pitching profiles.

reconstruction, which scouts weigh in their evaluation of injury risk and projection.

- **Pitch quality beyond velocity.** Pipeline's report notes that Anderson's fastball "plays much better than its velocity (sitting 92–94 mph, touching 97) because it carries past bats up in the strike zone." Pitch movement, spin rate, and perceived velocity are unavailable in our feature set. Two pitchers with identical K/9 can have substantially different professional projections based on these characteristics, which scouts evaluate but we cannot.
- **Repertoire depth.** Anderson developed a high-spin mid-80s slider as his primary breaking pitch in 2025 and uses an effective mid-80s changeup with fade and sink as a tertiary offering. A four-pitch mix with three above-average offerings is a starter-pipeline profile; our model sees only outcomes (strikeouts, walks, hits allowed), not the underlying mix.

- **Postseason performance.** Anderson led LSU to the 2025 national championship, won Most Outstanding Player honors at the College World Series, and threw a 1–0 shutout in the finals—17 innings of one-run baseball across two starts on the largest stage in college baseball. The CWS final concludes approximately two weeks before the draft. Our season-aggregate features include these games but cannot weight them differently from regular-season starts; scouts weight them heavily, viewing postseason performance as a signal of competitive resilience.
- **Physical projection and delivery.** Pipeline notes "a quick arm and athletic delivery" and "projection remaining in his slender 6-foot-2 frame." Body composition, delivery mechanics, and remaining physical projection are core scouting variables. Our

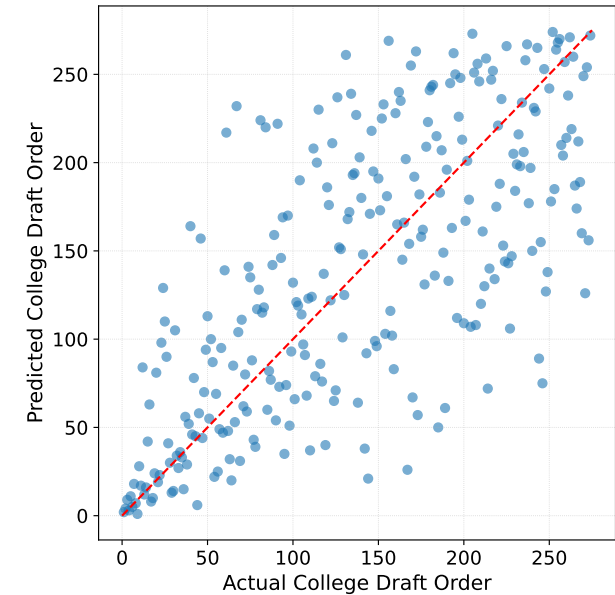


Figure 6. Predicted vs. actual college draft order, 2025 hold-out (n=274). Full class.

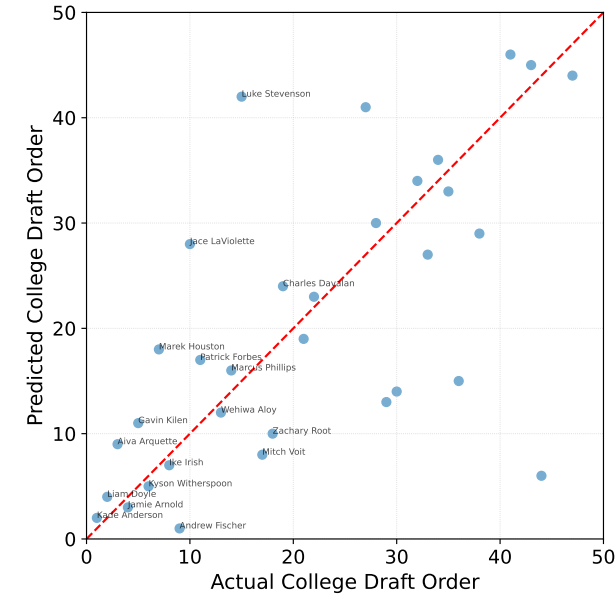


Figure 7. Predicted vs. actual college draft order, 2025 hold-out (n=274). Top-50 zoom with player labels. The model identifies the top-of-class draftees (Anderson, Doyle, Arnold, Arquette, Witherspoon) with strong accuracy.

model observes height and weight only for already-drafted players (Section 4), and observes nothing about delivery or projection at all.

The Anderson case illustrates the structural ceiling discussed in Sections 5.3 and 6. The model correctly identifies Anderson as a top-of-board pitcher in 2025—a notable success given that it operates only on publicly available season statistics. What it cannot do, by construction, is distinguish between “presumptive No. 1” and “presumptive top-3,” because the marginal information that resolves this—injury history weighted by recovery trajectory, pitch movement and spin, repertoire mix, postseason composite, and physical

Table 4. Top 20 actual college draftees in 2025: predicted draft order from the Stage 2 model. Strong agreement at the top of the board; mid-first-round misses (Bodine, Summerhill, Quick, LaViolette, Stevenson) reflect physical-tools and scouting signal not available to a stats-only model.

Actual	Player	Predicted	Δ
1	Kade Anderson	2	+1
2	Liam Doyle	4	+2
3	Aiva Arquette	9	+6
4	Jamie Arnold	3	−1
5	Gavin Kilen	11	+6
6	Kyson Witherspoon	5	−1
7	Marek Houston	18	+11
8	Ike Irish	7	−1
9	Andrew Fischer	1	−8
10	Jace LaViolette	28	+18
11	Patrick Forbes	17	+6
12	Caden Bodine	84	+72
13	Wehiwa Aloy	12	−1
14	Marcus Phillips	16	+2
15	Luke Stevenson	42	+27
16	Riley Quick	63	+47
17	Mitch Voit	8	−9
18	Zachary Root	10	−8
19	Charles Davalan	24	+5
20	Brendan Summerhill	81	+61

Table 5. Per-year head-to-head against MLB Pipeline on matched college players (2021–2025). The model is competitive with Pipeline in stable years (2024, 2025) but underperforms in 2021 (the first post-pandemic draft, with limited prior-year statistical signal). Mean across years computed over the per-year values.

Year	n	Model ρ	Pipe. ρ	Model MAE	Pipe. MAE
2021	99	0.468	0.822	23.1	13.0
2022	97	0.659	0.878	18.5	10.6
2023	108	0.605	0.801	20.9	14.2
2024	96	0.688	0.864	17.2	11.0
2025	97	0.665	0.838	16.8	11.3
Mean	497	0.617	0.841	19.3	12.0

projection—is not present in any publicly available season-aggregate dataset.

Anderson’s scouting report. The model’s predictions for Anderson, paired with the scouting report layer described in Section 4.5, are summarized in Table 6. The model assigns a Draft Grade of A+ ($P[\text{draft}] = 99.9\%$, 98th percentile of training-set predictions) and a Rank Grade of A+ (predicted pick ~3, 100th percentile). The one-sentence narrative reads: “The model sees a top profile (~pick 3). Driven by SO (180 vs. 55 median). Main red flag: Age (20 vs. 21 median).” The Stage 3 model predicts a signing bonus of \$4.17M against a predicted slot value of \$4.68M (bonus/slot ratio 0.89), implying the team will likely sign Anderson modestly below the slot value he commands.

Three observations about Anderson’s report are worth highlighting. First, the model identifies strikeouts as Anderson’s defining strength (raw count, K%, K-BB%, team-context strikeouts, and K/9 occupy all five strength slots). This matches both his statistical profile (180 strikeouts in 119 innings led NCAA Division I in 2025) and the

Table 6. Scouting report for Kade Anderson (actual draft pick: #3 overall). Draft probability is on the percentage-point scale; importance $\times z$ is the diagnostic cross-check. The model identifies strikeout production as the dominant strength and a slight age effect as the only meaningful concern.

Feature	Value	Median	$\Delta P[\text{draft}]$	Imp $\times z$
<i>Top 5 strengths</i>				
SO	180.0	55.0	+0.26%	+0.199
K%	0.374	0.203	+0.16%	+0.044
K-BB%	0.301	0.114	+0.10%	+0.026
SO.team	760.0	432.0	+0.06%	+0.018
K/9	13.6	8.20	+0.05%	+0.039
<i>Top 5 concerns</i>				
Age	20.0	21.0	−0.05%	−0.053
G (pitch)	19.0	15.0	−0.02%	+0.004
A.team	543.0	484.0	−0.02%	+0.005
Home_WP.team	0.854	0.607	−0.01%	+0.007
SB.team	51.0	55.0	−0.01%	−0.001

scouting consensus, which also emphasizes his swing-and-miss ability. Second, the listed concerns are quantitatively small ($|\Delta P[\text{draft}]| \leq 0.05\%$), indicating that the model's prediction is dominated by the strengths rather than dragged down by competing concerns—an internally consistent A+ profile. Third, the model flags Anderson's age (20, one year below the modal junior age of 21) as a “concern,” which is somewhat counterintuitive: younger players typically command a premium. This artifact arises because the model captures the non-monotonic age effect—draft probability is maximized at age 21 (the modal junior), so being 20 is a marginal departure from the peak—without reasoning over the long-horizon development implications that scouts apply. We address this in Section 6.

Side-by-side with MLB Pipeline. Pipeline's pre-draft writeup (MLB Pipeline 2021-2025) documents several facts not encoded in any of our 92 features: “Anderson had Tommy John surgery in April 2022 and missed his Louisiana high school senior season”; “Anderson's fastball plays much better than its velocity (sitting 92–94 mph, touching 97)”; “a quick arm and athletic delivery”; and his 17-inning, one-run College World Series performance two weeks before the draft. The scouting report makes explicit, in structured form, exactly which channels of the prediction the model is and is not informed by. For Anderson, the model captures the strikeout-based signal that anchors both evaluations but cannot integrate injury history, pitch movement, delivery quality, or postseason composite weighting.

This observation has two practical implications. For player decision support, the model's predicted rank should be interpreted as a stats-implied projection that may be refined upward by scouting evaluation: a player with Anderson's statistical profile is at least a top-3 college pick by statistics alone, and scouting can move them up but rarely sharply down. For team analytics, the case illustrates that the marginal value of in-person scouting at the very top of the draft is small in absolute rank terms (one or two picks) but large in signing-bonus and franchise-cornerstone terms.

Table 7. Top 10 features for the Stage 3 signing-bonus regressor (gain-based).

Feature	Gain
age	56.6
wrc.bat	16.9
so_pitch	15.0
r.bat	12.3
wraa.bat	12.0
conf_rpi.team	11.9
conf_national.wp.team	11.7
hr.bat	11.7
k_pitch	11.0
k-bb_pitch	9.7

5.5 Stage 3: Signing bonus

The Stage 3 regressor achieves MAE of \$370,586, $R^2 = 0.454$, and Spearman $\rho = 0.731$ on the 2025 hold-out ($n=273$). The Spearman exceeds the multi-year head-to-head benchmark Spearman of 0.617 (Table 5), indicating that signing bonus—which incorporates both team valuation and player negotiating leverage—is in some respects a more predictable quantity than raw draft order against MLB Pipeline.

Feature importance (Table 7) overlaps substantially with Stages 1 and 2: age, wRC, strikeout metrics, and conference RPI all appear in the top 10. The new RPI variables make a modest appearance here (SOS.team at rank 12, rpi.team at rank 15), suggesting that the strength-of-schedule signal that mattered most for Stage 1 is partially redundant with conference RPI for predicting bonus.

Figure 8 shows predicted versus actual signing bonus on the log scale. The model systematically under-predicts in the extreme high tail: for the top 2025 pick (Anderson, actual \$8.8M), the predicted bonus was \$3.5M—a 60% underestimate. This pattern is characteristic of log-transformed regression on right-skewed targets: the log transform stabilizes variance during training but compresses the extreme upper tail, so the back-transformed model regresses toward the conditional median of the dollar distribution rather than its conditional mean. The effect is concentrated in the top ~10 picks of each draft, where actual bonuses range from \$3M to \$10M+ but the model rarely exceeds \$4M. For the bulk of the distribution (\$100K–\$2M, comprising roughly 80% of drafted players), predictions are well-calibrated.

The operational consequence is that the Stage 3 model is informative for rank-order purposes ($\rho = 0.731$ confirms this) but should not be used as a point estimate of dollar value for players projected near the top of the draft. We discuss two mitigations as future work (Section 7): (i) a quantile-regression variant that targets specific percentiles of the bonus distribution rather than the conditional mean, and (ii) a two-stage approach in which a separate top-tail model is fit on players with predicted log-bonus above a threshold and used to correct the primary model's output. For the present analysis, we report dollar predictions for top-of-draft players as a lower bound rather than a point estimate.

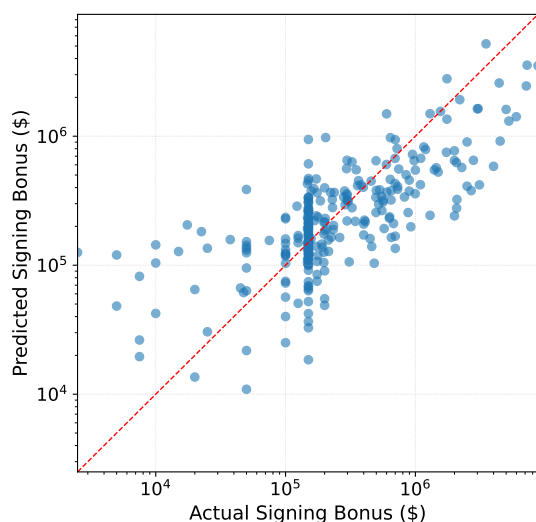


Figure 8. Predicted vs. actual signing bonus on log scale, 2025 hold-out ($n=273$). The 45° line indicates perfect prediction. The model under-predicts in the extreme top tail (top-10 picks) due to log-transform regression-to-mean dynamics; predictions in the \$100K–\$2M range are well-calibrated.

5.6 Cross-model feature consistency

The three stages predict distinct quantities (draft probability, draft order conditional on being drafted, signing bonus conditional on being drafted) but converge on a similar set of dominant features. Table 8 summarizes the top-10 feature sets across the three stages. Four features appear in all three top-10 lists: `age`, `wrc_bat`, `so_pitch`, and `conf_rpi_team`. This convergence supports an interpretation in which draft decisions, draft-order ranking, and bonus negotiation all reflect a common underlying signal: a player’s combined offensive production and strikeout ability, conditioned on age and competitive context.

The team RPI features (`rpi_team`, `SOS_team`) make their strongest appearance in Stage 1, where `rpi_team` ranks second overall. In Stages 2 and 3, both features fall outside the top 10 (`rpi_team` ranks 27 in Stage 2, 15 in Stage 3; `SOS_team` ranks 16 in Stage 2, 12 in Stage 3). The pattern suggests that team strength of schedule contributes most to the binary draft-or-not decision—where scouts use a player’s team context as a key heuristic—and less to ordinal placement and dollar value, which depend more directly on individual performance.

5.7 Scouting reports: additional cases

Anderson’s report in Section 5.4 demonstrates the system at the elite end. To show that the scouting report layer behaves coherently across the full range of profiles—and that the suppression rule introduced in Section 4.5 works in practice—we present a clearly non-draftable profile (Table 9).

Cooper Clark is a right-handed pitcher at Campbell University in 2024 with a 7.71 ERA across 51.1 innings, a 1.58 WHIP, and 47 strikeouts. The classifier produces $P[\text{draft}] = 0.1\%$ (an F grade); because this is below the 25% suppression threshold, the report hides the rank and bonus predictions entirely rather than display unreliable extrapolations.

Table 8. Top-10 features by gain across the three pipeline stages. Bold features appear in all three stages’ top-10. A dash indicates the feature does not appear in that stage’s top 10.

Feature	Stage 1	Stage 2	Stage 3
age	1	2	1
wrc_bat	3	3	2
so_pitch	5	5	3
conf_rpi_team	9	6	6
<code>rpi_team</code>	2	—	—
<code>slg_bat</code>	4	—	—
<code>SOS_team</code>	6	—	—
<code>r_bat</code>	7	—	4
<code>h_bat</code>	8	—	—
<code>ops_bat</code>	10	—	—
<code>k-bb%_pitch</code>	—	1	10
<code>SO_team</code>	—	4	—
<code>K/9_team</code>	—	7	—
<code>conf_national_wp_team</code>	—	8	7
<code>wraa_bat</code>	—	9	5
<code>bb/k_bat</code>	—	10	—
<code>hr_bat</code>	—	—	8
<code>k%_pitch</code>	—	—	9

Table 9. Scouting report for Cooper Clark (Campbell University, 2024). The classifier assigns $P[\text{draft}] = 0.1\%$ (F grade), and the rank and bonus predictions are suppressed by the 25% threshold rule. The report identifies the strongest available concerns rather than fabricating downstream numbers.

Feature	Value	Median	$\Delta P[\text{draft}]$	$\text{Imp} \times z$
<i>Top 5 strengths (note small magnitude)</i>				
ERA-FIP	2.159	−0.363	+0.02%	+0.010
WP	8.0	5.0	+0.01%	+0.006
Neutral_WP_team	1.000	0.500	+0.01%	+0.008
NC_Rec_WP_team	0.731	0.500	+0.01%	+0.012
G (pitch)	13.0	15.0	+0.01%	−0.002
<i>Top 5 concerns</i>				
SO	47.0	55.0	−0.03%	−0.013
DPPG_team	0.470	0.660	−0.02%	−0.007
K%	0.194	0.203	−0.01%	−0.002
SOS_team	88.0	146.0	−0.01%	−0.019
SBPG_team	1.850	1.020	−0.01%	+0.009

Three structural features of the report are visible here. First, the strength entries have very small magnitudes ($\Delta P[\text{draft}] \leq 0.02\%$): when the overall draft probability is essentially zero, no feature can be a meaningful “strength” in absolute terms. The report nonetheless surfaces the least-negative features so the player can see which aspects of their profile are not actively hurting them. Notably, two of Clark’s five surfaced “strengths” are team-context wins-percentages (Neutral_WP, NC_Rec_WP)—a clean illustration of how the team-RPI feature group propagates through to the individual-player report.

Second, the concerns are dominated by individual strikeout deficits (raw SO and K%) and team-context weakness (DPPG, low SOS, high SBPG against). Of these, SO and K% are addressable through development; SOS reflects program selection (Campbell competes in the CAA, ranked 88th by SOS against a training median of 146); DPPG and SBPG reflect team defensive context that an individual player cannot directly improve. The report makes this distinction visible: a player can see which of their

concerns are personal development targets versus structural team factors.

Third, the suppression rule prevents the user from being presented with a confident-looking “predicted pick 350” or “predicted signing bonus \$10K,” both of which would be extrapolations outside the rank and bonus models’ training distributions (drafted players only).

The Clark report is operationally important: a college pitcher considering the draft can run the report on their own statistics and receive either (a) a coherent assessment of their stock with grades and explanations, or (b) a clear “the model does not see this as a draftable profile” signal with no false specificity below it. The system’s behavior in the high-probability regime (Anderson) and the low-probability regime (Clark) are mutually consistent: in both cases, the report makes the underlying model behavior interpretable to a non-technical user.

6 Discussion

6.1 Practical implications

The pipeline supports two distinct decision contexts.

For NCAA juniors weighing draft entry against returning to school, the combination of Stage 1 (am I likely to be drafted?), Stage 2 (where am I likely to fall?), and Stage 3 (what bonus can I expect?) provides a joint distribution over outcomes that informs the leverage calculation underpinning the return-to-school decision. Crucially, the scouting report layer (Section 4.5) converts these three numbers into player-actionable feedback. A junior contemplating their options can run their season’s statistics through the system and receive: a draft letter grade, a predicted rank range, a predicted bonus range, and—most actionably—a structured list of which statistics are helping their draft stock and which are hurting it. Strength and concern entries name specific on-field statistics with the player’s value compared to the training-set median (e.g., “K/9 = 9.8 vs 8.2 median”), giving concrete development targets. A pitcher whose top concern is BB/9 has a clear off-season focus; a batter whose top concern is K% (bat) has a clear plate-discipline target.

The suppression rule (Section 4.5) is also operationally important: a player with a clearly-undrafted profile receives an F grade and explicit acknowledgment that the rank and bonus predictions are unreliable for them, rather than confident-looking extrapolations like “predicted pick 350.” This protects the system’s credibility and prevents misuse.

For team analytics groups, the gap between the stats-only model and MLB Pipeline (Section 5.3) quantifies the marginal predictive value of in-person scouting over publicly available statistics. The 0.22 Spearman gap on draft order corresponds to approximately seven additional ranks of MAE—a meaningful but bounded improvement, useful for calibrating analytics-vs.-scouting resource allocation. The scouting report further supports internal communication: an analyst can share a one-page report on a non-top-of-board player with scouts or front-office personnel without requiring them to inspect the underlying feature matrix.

6.2 Why age dominates

Age appears as the top feature in all three stages by a wide margin. We interpret this through two channels. *Mechanically*, older players have more seasons of NCAA performance, increasing the signal density per player and reducing the variance of derived statistics. *Substantively*, teams place a premium on younger players for two reasons: development upside (a 20-year-old has more remaining peak years than a 22-year-old) and leverage (a junior retains return-to-school option value, while a senior does not, shifting bargaining power in the team’s favor). The non-monotonic age PDP (Figure 3) is consistent with both interpretations: the peak at age 21 reflects the modal college junior, and the post-21 decline reflects the leverage and upside considerations that disadvantage seniors.

The non-monotonicity also produces the apparent “age concern” for Anderson in Table 6: at age 20, his SHAP attribution for age is slightly negative because the model has learned that age 21 is the local mode of the draft-probability distribution. This is technically correct as a description of the model’s behavior on the 2021–2025 cohort, but it would be misleading as career advice: a 20-year-old with elite statistics has more remaining development time than a 21-year-old with the same statistics. The report’s small magnitude on this entry (−0.05% on a 99.9% baseline) limits its practical importance, but the artifact illustrates a general caution: the report describes what the model is doing, not what is normatively true.

6.3 Strikeouts dominate pitching predictors

Strikeout metrics (`so_pitch`, `k%`, `k-bb%`, `team K/9`) appear consistently across all three stages. This aligns with the modern scouting emphasis on swing-and-miss ability, which is valued in part because strikeouts are defense-independent (McCracken 2001): a pitcher generating strikeouts records outs that do not depend on the fielders behind him. This is the foundational insight of defense-independent pitching statistics, later operationalized in metrics such as FIP, which we include as a feature. This is particularly relevant for projecting college pitchers into professional environments, where defensive quality differs systematically from the NCAA baseline.

6.4 Limitations

Data scope. The 2021–2025 window controls for equipment and draft-rules regimes but is short. Year-effects (such as the 2021 anomaly identified in Section 5.3) cannot be modeled with only five years of data.

Population. The pipeline applies to NCAA college players only. High-school draftees, junior-college players, and international amateurs are excluded.

Feature coverage. The model uses statistical performance and limited team context only. It does not observe velocity, exit velocity, sprint speed, or other physical-tools measurements that scouts use heavily. Physical attributes (height, weight, handedness, position) are observed only for drafted players via the MLB Stats API and so cannot be used in Stage 1.

Imbalance handling. Random undersampling for Stage 1 discards majority-class information. A complete sensitivity

analysis comparing undersampling, `scale_pos_weight`, and SMOTE is left to future work.

Hold-out size. Stages 2 and 3 are evaluated on a single year (2025; $n=274$ and 273 respectively). The multi-year head-to-head for Stage 2 partially addresses this, but Stage 3 robustness across years is not directly demonstrated.

Scouting report attributions are local explanations, not causal claims. SHAP attributions describe the marginal contribution of each feature to the model's prediction given the other features observed for that player. They do not establish that changing a feature would change real-world draft outcomes. A pitcher whose top concern is BB/9 is told that high walk rate is hurting their model-predicted draft stock; whether reducing walks would also reduce their actual draft position is not directly testable from this analysis. The artifacts in the Anderson age attribution (Table 6) illustrate this distinction: the model captures a non-monotonic age effect that does not translate to a career-advice recommendation.

Structural ceiling on draft-order prediction. This is the most important limitation. Pre-draft ranking and actual draft pick number diverge for non-statistical reasons: bonus demands (players seeking above-slot bonuses fall to teams willing to pay), team strategy (underslot and overslot trades to bank pool money across rounds), positional preferences (teams with positional needs reach for players at scarce positions), signability (some players are committed to returning to school absent specific bonus thresholds), and medical concerns (undisclosed injuries cause downward movement). None of these signals are available in on-field statistics. The implication is that the attainable Spearman against actual draft order is bounded above by some value strictly less than 1.0, even for a hypothetical perfect-information ranker. Our 0.710 Spearman should be interpreted against this implicit ceiling rather than against unity.

6.5 Threats to validity

Look-ahead concerns. NCAA statistics in our feature matrix are aggregated over the full season, including games that conclude shortly before the draft. The College World Series runs through late June and the draft is held in mid-July, so the most consequential late-season games—super-regionals and the CWS itself—are temporally co-located with the period during which professional scouts finalize their evaluations. For top prospects whose CWS performance materially affected their pre-draft stock (the Anderson case study in Section 5.4 is an example), our season-aggregate features include this performance and may therefore reflect signal that scouts also incorporated into their final rankings. Two factors bound the effect. First, the bulk of the season's PA and IP occur well before draft week; for most players, $> 90\%$ of their statistical signal is generated by mid-May. Second, our model does not differentially weight postseason games (Section 5.4), so any look-ahead effect is diluted by the much larger regular-season contribution. We have not quantified the effect by retraining with games-before-June-1 cutoff features, and this remains a residual threat to internal validity.

Selection bias from API matching. The 5 fuzzy-name-match drops (0.3% of drafted players) and any drafted

players who never appeared in the NCAA player-statistics scrape (Section 3) represent a small but non-random exclusion. We estimate the unmatched fraction to be on the order of a few percent of the total drafted population, concentrated in lower-volume contributors (low PA/IP) and in players whose name spellings differ substantially between NCAA records and the MLB Stats API. Stage 1 is unaffected by API matching (it uses NCAA data only); Stages 2 and 3 condition on having been drafted and matched, so their predictions are evaluated only on the matched subset and the bias is one of population scope rather than measurement error.

Hyperparameter tuning and test-set leakage. The single-year hold-out hyperparameters were chosen by inspecting validation MAE on 2024 with training on 2021–2023. The same configuration is reused for the multi-year LOYO benchmark, which means the 2025 fold of the LOYO uses hyperparameters tuned on a validation set (2024) that overlaps the LOYO training data for the 2025 fold. This is a mild form of information bleed: a hyperparameter that happens to be good for 2024 is being used to predict 2025 with the benefit of having seen 2024 during tuning. We view this as small in practice—hyperparameters were chosen via coarse manual grid search and are not finely tuned—but it is non-zero.

7 Future work

We identify several priority extensions:

1. **Stage 3 calibration in the top tail.** The Stage 3 underprediction for top-of-draft players (Section 5) admits at least two principled fixes. A quantile-regression variant of XGBoost (using the `reg:quantileerror` objective) can target specific percentiles of the bonus distribution rather than the conditional mean, providing calibrated intervals directly. Alternatively, a two-stage approach can fit a top-tail model on players with predicted log-bonus above a threshold and use its output to correct the primary model. Either approach should be evaluated against the current log-target regression on multi-year hold-outs.
2. **Sensitivity analysis on fuzzy-match threshold.** The 70% similarity threshold for player-to-API matching (Section 3) was chosen heuristically. A systematic sensitivity analysis at thresholds in $\{0.50, 0.60, 0.70, 0.80, 0.90\}$, combined with manual adjudication of the marginal cases, would quantify the robustness of the drafted-subset population and could be extended to a learned matching procedure if the manual labels prove informative.
3. **Outcome-based evaluation.** Construct a reproducible developmental-success benchmark using frozen post-draft snapshots taken at a fixed calendar date for multiple cohorts, with sufficient time elapsed for development outcomes to be informative. This would directly address the structural ceiling on draft-order evaluation discussed above.
4. **Multi-year out-of-time validation for Stages 1 and 3.** As 2026 and 2027 draft data accrue, rerun the pipeline with rolling-window validation to

characterize year-to-year stability and detect drift in feature importance.

5. **Physical-attribute integration for the full dataset.** Currently height, weight, and position are observed only for drafted players. Integration with publicly available NCAA roster data (or with Perfect Game amateur scouting profiles, where applicable) would extend these features to non-drafted players and allow them to be used in Stage 1.
6. **Counterfactual recommendations in the scouting report.** The current report identifies which features are hurting a player's draft stock but does not estimate what the player's grade would become if a specific concern were addressed (e.g., "reducing BB/9 from 4.5 to 3.0 would raise predicted rank by approximately 30 ranks"). Extending the report with counterfactual queries against the trained models is a natural next step.
7. **User study of the scouting report layer.** The scouting report is currently evaluated only through the structural sanity-checks in Section 5.7. A user study with NCAA players, college coaches, and player agents would test whether the report's grades, narratives, and feature attributions are interpretable and actionable for the intended audience, and would surface terminology or framing changes needed for non-technical use.
8. **High-school draftees.** The current pipeline excludes high-school players. A parallel pipeline for HS draftees is methodologically straightforward but requires a separate data infrastructure.
9. **Integration with the Gerber competing-risks framework (Gerber 2023).** The three stages of our pipeline produce outputs (draft probability, draft order, signing bonus) that could serve as inputs to a downstream model predicting MLB debut, time-to-debut, and retirement risk, creating an end-to-end amateur-to-MLB career-value estimator.

8 Conclusion

This paper presents an end-to-end pipeline for pre-draft prediction of MLB draft outcomes from NCAA player statistics, consisting of a binary draft classifier (89.1% balanced accuracy; PR-AUC 0.687 on the operational imbalanced distribution against a 0.086 baseline, ROC-AUC 0.956), a college draft-order regressor (Spearman 0.710, MAE 46.5 ranks on 2025 hold-out), a signing-bonus regressor (Spearman 0.731, MAE \$371K, $R^2 = 0.454$ on 2025 hold-out, with acknowledged underprediction in the extreme top tail), and a SHAP-based scouting report layer that translates these predictions into letter grades, narrative summaries, and ranked strengths and concerns at the individual-player level. A second contribution is a five-year head-to-head benchmark against MLB Pipeline's pre-draft top-250 rankings (n=497 matched college players, 2021–2025), which quantifies the marginal predictive value of in-person scouting beyond publicly available statistics at a mean Spearman gap of 0.224 and 7.3 ranks of MAE. The gap is partially structural (signability, bonus demands, team strategy create irreducible noise in actual pick number)

and partially substantive (Pipeline incorporates physical-tools evaluation, pitch quality, and team-internal signaling that statistical features cannot proxy). Across all three predictive stages, age, team strength of schedule (RPI, SOS), offensive production (wRC, OPS, SLG), and strikeout ability (SO, K-BB%) emerge as dominant predictors. By pairing the predictive models with structured per-player feedback, the pipeline supports NCAA players evaluating draft entry against return-to-school, and provides team analytics groups a calibrated estimate of the predictive value scouts add over public data.

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References

- Baseball Almanac (2021-2025) Baseball draft history. URL https://www.baseball-almanac.com/draft/baseball_draft.shtml.
- Biggs M (2025) ncaa_bbStats. URL <https://github.com/CodeMateo15/CollegeBaseballStatsPackage>.
- Chawla NV, Bowyer KW, Hall LO and Kegelmeyer WP (2002) SMOTE: Synthetic minority over-sampling technique. *J. Artif. Intell. Res.* 16: 321–357.
- Chen T and Guestrin C (2016) XGBoost: A scalable tree boosting system. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, KDD '16. ACM, pp. 785–794. DOI:10.1145/2939672.2939785.
- Davenport C and Woolner K (1999) Revisiting the pythagorean theorem. *Baseball Prospectus*.
- FanGraphs (2021-2025) College leaderboards. URL <https://www.fangraphs.com/leaders/college>.
- Gerber E (2023) Competing risks analysis of mlb draft data. Poster. URL <https://e-gerber.shinyapps.io/DraftSurvival/>. DOI: 10.13140/RG.2.2.28623.46249.
- Huang ML and Li YZ (2021) Use of machine learning and deep learning to predict the outcomes of major league baseball matches. *Applied Sciences* 11(10).
- Lundberg SM and Lee SI (2017) A unified approach to interpreting model predictions. In: *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.
- Major League Baseball Advanced Media (2021-2025) MLB stats API. URL <https://statsapi.mlb.com/>. Endpoints used: /api/v1/draft/{year}. Accessed 2026.
- McCracken V (2001) Pitching and Defense: How Much Control Do Hurlers Have? — *Baseball Prospectus* — [baseballprospectus.com](https://www.baseballprospectus.com/news/article/878/pitching-and-defense-how-much-control-do-hurlers-have). <https://www.baseballprospectus.com/news/article/878/pitching-and-defense-how-much-control-do-hurlers-have> [Accessed 09-06-2026].
- MLB Pipeline (2021-2025) MLB draft prospect rankings. URL <https://www.mlb.com/prospects/draft/>. Top-250 pre-draft prospect rankings scraped from per-year pages

- (/mlb/prospects/{year}/draft/) for the 2021–2025 draft classes.
- NCAA (2021-2025) Ncaa statistics. URL <https://stats.ncaa.org/>.
- Nolan W (2021-2025) RPI (live) rankings for college baseball. URL <https://www.warrennolan.com/baseball/>. Conference, non-conference, home, road, neutral, and Q1–Q4 win-percentage splits accessed for the 2021–2025 seasons.
- Penn M (2023) The value of playing in college baseball to major league baseball organizations .
- Russell DA (2005) Non-wood bats: Performance trends in college baseball. Graduate Program in Acoustics, The Pennsylvania State University.
- Saito T and Rehmsmeier M (2015) The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS One* 10(3): e0118432.
- Seneker D (2025) Maximizing the MLB draft: How college performance and leverage impact player signing bonus and team payoff. *CMC Senior Theses* .
- Zhao W, Akella VS, Yang S and Luo X (2025) Machine learning in baseball analytics: Sabermetrics and beyond. *Information* .